TESS planet candidate around TIC 219396153

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Summary

Signal found by the bls search of a public-data TESS pipeline (SPOC 120 s PDCSAP light curves, sectors 87). Final tier: **candidate: likely planet**. Catalogue cross-match (TOIs, CTOIs, published planets, SPOC TCEs, TESS eclipsing binaries): **TOI 7888.01** (TOI). CNN score (calibrated): 0.928. TRICERATOPS (10 Monte Carlo runs, mean ± error of the mean): FPP = 0.155 ± 0.0069, NFPP = 0.

Host star (TIC v8)

RA, Dec (deg)	77.838671, -49.357726
TESS magnitude	13.6
Radius ($R_\odot$)	0.642 ± 0.0193
$T_{\rm eff}$ (K)	3859 ± 157
Density ($\rho_\odot$)	2.35 ± 0.135
Distance (pc)	220

Transit fit

Limb-darkened model (quadratic law, Kipping q_1, q_2; GPU float64, 1024 annuli, 1024x4096 tables, 15x supersampling), emcee with 48 walkers, 12000 steps (802 burn-in), run until the chain exceeded 50 autocorrelation times. Medians and 68% intervals.

Mid-transit time T_0 ($\mathrm{BJD_{TDB}}$)	$2460668.7548^{+0.0036}_{-0.0035}$
Period P (days)	$10.2370^{+0.0038}_{-0.0034}$
Radius ratio $R_p/R_\star$	$0.0950^{+0.0096}_{-0.0074}$
Impact parameter b	$0.49^{+0.28}_{-0.32}$
Stellar density $\rho_\star$ (g cm^{-3})	$2.2^{+1.2}_{-1.3}$
Limb darkening q_1	$0.50^{+0.32}_{-0.30}$
Limb darkening q_2	$0.48^{+0.33}_{-0.31}$
Depth (ppm)	10518^{+1129}_{-1091}
Duration T_{14} (hours)	$3.32^{+0.33}_{-0.25}$
$a/R_\star$	$23.0^{+3.5}_{-5.6}$
Inclination (deg)	$88.8^{+0.8}_{-1.3}$
Planet radius ($R_\oplus$)	$6.66^{+0.71}_{-0.56}$
Semi-major axis (au)	$0.069^{+0.011}_{-0.017}$
Equilibrium temperature (K, $A = 0$)	572^{+84}_{-48}
Insolation ($S_\oplus$)	18^{+13}_{-5}

Vetting

Test	Value	Pass condition	Result
Transit SNR (correlated noise)	9.63	≥ 7.1	pass
Transits observed	2	≥ 2 (periodic)	pass
Odd/even depth difference (σ)	–	≤ 4	pass
Secondary eclipse (σ)	2.59	≤ 3.9	pass
Ingress fraction (V shape)	0.242	≤ 0.4	pass
Planet radius ($R_\oplus$)	7.19	≤ 25	pass
Centroid source offset (arcsec)	4.21	4σ and $>21''$	pass
Difference image offset (arcsec)	–	4σ and $>10.5''$	pass
Difference image significance (σ)	–	–	–
CCD common mode (σ)	-0.542	≤ 4	pass
Crowding (CROWDSAP)	0.994	≥ 0.8	pass
Stellar density ratio (max/TIC)	0.974	≥ 0.2	pass

Caveats

- This is a candidate with statistical evidence, not a confirmed planet: confirmation needs high-resolution imaging, reconnaissance spectroscopy and radial velocities.
- TRICERATOPS was run without a high-resolution imaging contrast curve, so close bound companions are not excluded.
- The MCMC converged (chain longer than 50 autocorrelation times, stable estimate) (acceptance 0.18, 12000 steps, about 3342 independent samples); errors are scaled by the time-averaging factor $\beta = 1.18$ for correlated noise.
- Stellar radius and effective temperature come from TIC v8; if the TIC gives no error, 7% (radius) and 150 K were assumed.

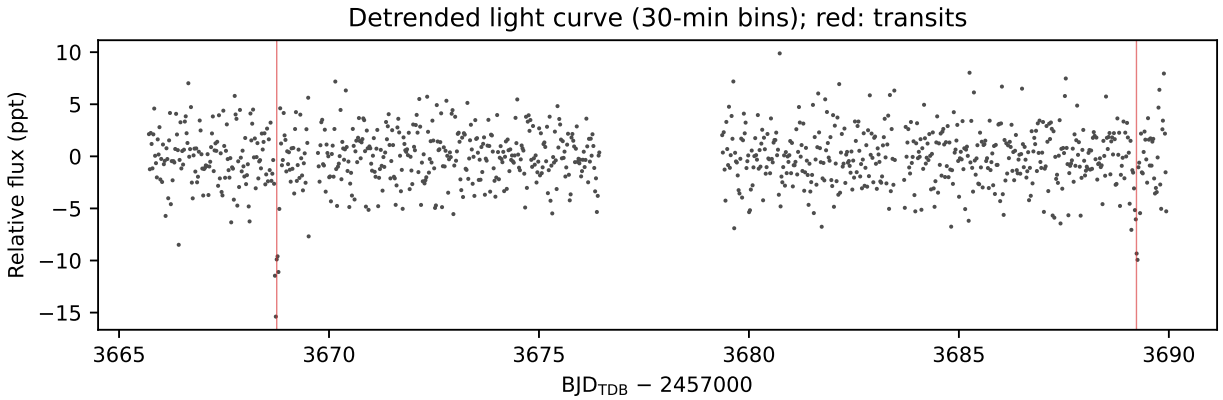


Figure 1: Detrended SPOC PDCSAP light curve (transit-protected biweight).

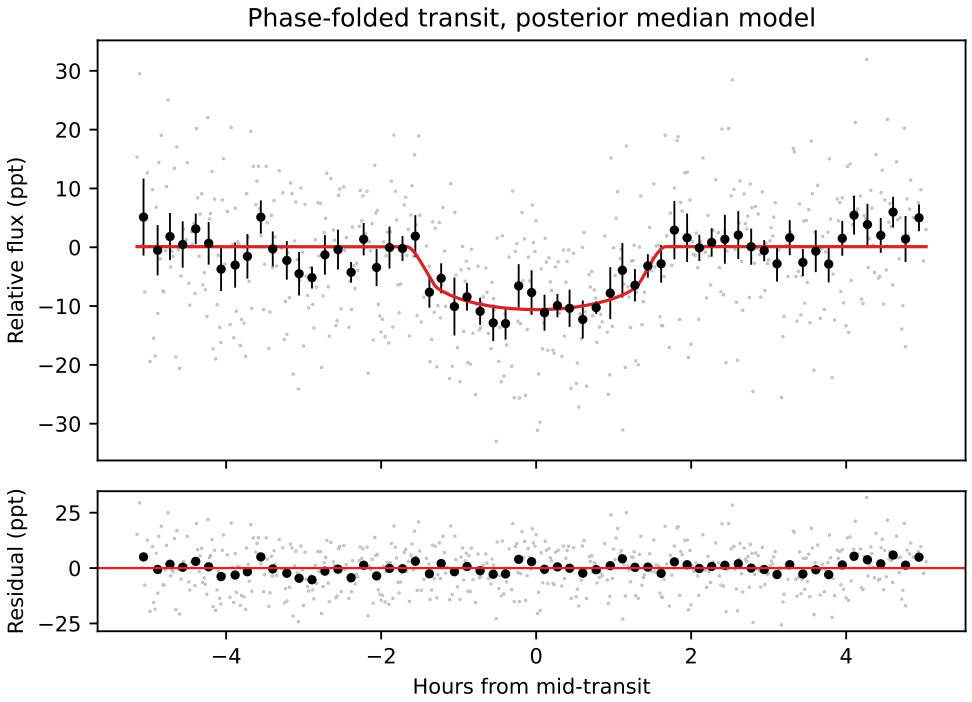


Figure 2: Phase-folded transit with the posterior median limb-darkened model.

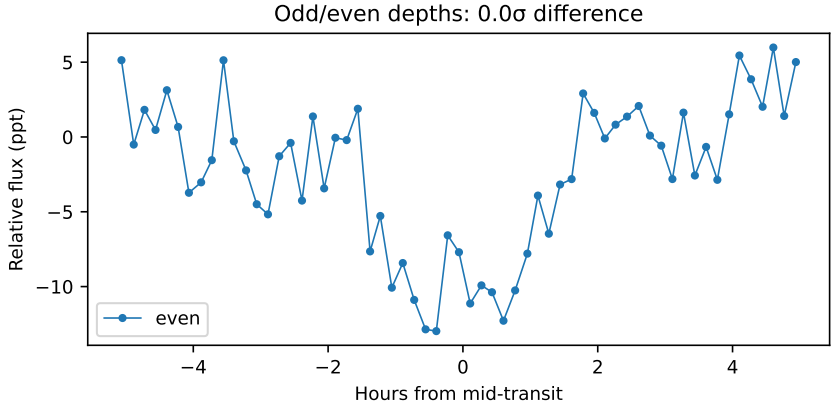


Figure 3: Odd and even transits, binned to 10 minutes.

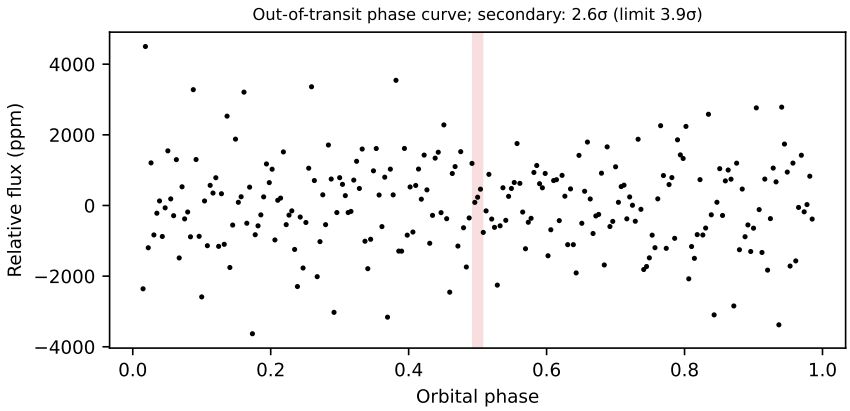


Figure 4: Phase curve; the shaded band is the secondary eclipse window for a circular orbit.

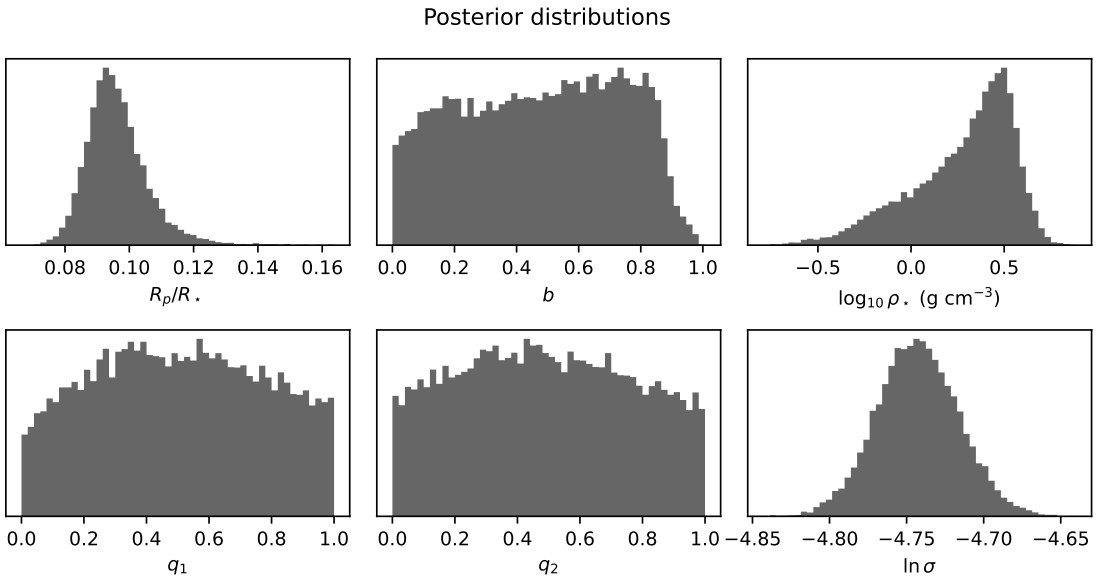


Figure 5: Posterior distributions of the fitted parameters.